

- (5.4.) 6. Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be a basis for a vector space V . Find $T(2\mathbf{b}_1 - \mathbf{b}_2 + 4\mathbf{b}_3)$ when T is a linear transformation from V to V whose matrix relative to $\mathcal{B}$ is

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & -6 & 1 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix}.$$

Sol. Recall that each column of $[T]_{\mathcal{B}}$ is the $\mathcal{B}$ coordinate of each $\mathbf{b}$ vectors in order. For example, the first column $(0, 0, 1)$ means $T(\mathbf{b}_1) = 0 + 0 + \mathbf{b}_3$.

We can first obtain the $\mathcal{B}$ coordinate of $T(2\mathbf{b}_1 - \mathbf{b}_2 + 4\mathbf{b}_3)$ by

$$\begin{bmatrix} 0 & -6 & 1 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ -9 \\ 32 \end{bmatrix}.$$

Notice that the vector $(2, -1, 4)$ came from reading $2\mathbf{b}_1 - \mathbf{b}_2 + 4\mathbf{b}_3$ in $\mathcal{B}$ coordinate. So we are keeping everything in the same ($\mathcal{B}$) coordinate, and it is merely a linear transformation done in this new coordinate until now.

Now the remaining job is reading the resulting vector $(10, -9, 32)$, which is in a non-standard coordinate. Read it as a linear combination of $\mathbf{b}$ vectors as $10\mathbf{b}_1 - 9\mathbf{b}_2 + 32\mathbf{b}_3$.

8. Find the $\mathcal{B}$ matrix for the transformation $\mathbf{x} \mapsto A\mathbf{x}$, when $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$.

$$A = \begin{bmatrix} -1 & 4 \\ -2 & 3 \end{bmatrix}, \mathbf{b}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Sol. Recall the definition of $\mathcal{B}$ matrix of a linear transformation. Find $A\mathbf{b}_1$ and $A\mathbf{b}_2$ using matrix-vector multiplication first, as

$$A\mathbf{b}_1 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, A\mathbf{b}_2 = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

Then read the vectors in the $\mathcal{B}$ coordinate to build the columns of the $\mathcal{B}$ matrix of A . By inspection, $(5, 0) = \mathbf{b}_1 - 2\mathbf{b}_2$ and $(5, 5) = 2\mathbf{b}_1 + \mathbf{b}_2$. Or you could augment the matrix $[\mathbf{b}_1 \ \mathbf{b}_2]$ by $\begin{bmatrix} 5 & 5 \\ 0 & 5 \end{bmatrix}$ and make it to RREF to pick up the resulting two columns on the right.

$$[[A\mathbf{b}_1]_{\mathcal{B}} \quad [A\mathbf{b}_2]_{\mathcal{B}}] = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}.$$

9. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$. Find a basis $\mathcal{B}$ for $\mathbb{R}^2$ with the property that $[T]_{\mathcal{B}}$ is diagonal.

$$A = \begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}.$$

Sol. This problem reduces to diagonalizing A using the Theorem 8 of 5.4. Find the eigenvalues by the characteristic polynomial

$$p_A(\lambda) = \begin{vmatrix} -\lambda & 1 \\ -3 & -\lambda + 4 \end{vmatrix} = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

The two roots of $p_A(\lambda)$ found as 1 and 3 are the eigenvalues of A . Now each eigenvalue will have a corresponding eigenspace, and choosing one nonzero eigenvector from each eigenspace, those two vectors will form the desired basis $\mathcal{B}$.

For $\lambda = 1$,

$$A - \lambda I = \begin{bmatrix} -1 & 1 \\ -3 & 3 \end{bmatrix}$$

clearly has $(1, 1)$ in its nullspace (meaning $(A - \lambda I)(1, 1) = 0$). The dimension of the eigenspace is 1 since the eigenvalue 1 has multiplicity 1. So $\{(1, 1)\}$ is a basis of this eigenspace.

For $\lambda = 3$,

$$A - \lambda I = \begin{bmatrix} -3 & 1 \\ -3 & 1 \end{bmatrix}$$

has $(1, 3)$ in its nullspace. Again with the same reasoning, $\{(1, 3)\}$ is a basis of this eigenspace.

Now let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

Then the Diagonal Matrix Representation Theorem (Theorem 8 mentioned earlier) guarantees that $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ is the $\mathcal{B}$ matrix of A .